



Department of Computer Science

Assignment#3

Course Code: CS-121

Course Title: Digital Logic Design

Total Marks: 15

Assigned Date: 21st May, 2025.

Name: _____

Registration No. _____

Discipline/Semester: BS(CS)- 2nd BSubmission Date/Deadline: 23th May, 2025.

Instructions:

- Late submissions will not be accepted.
- Hand written assignments will be accepted only

CLO3-GA4-BT-Level C4

Q1. **Implement** the following four Boolean expressions with three half adders.

- a) $D = A \oplus B \oplus C$
- b) $E = A'BC + AB'C$
- c) $F = ABC' + (A' + B')C$
- d) $G = ABC$

Q2: **Draw** the logic diagram of a 2-to-4-line decoder using

- (a) NOR gates only and
- (b) NAND gates only. Include an enable input.

Q3: **Implement** the BCD to Excess -3 conversion using K map.